Data Cleaning Process

Summary Report — Hospital Admission & Discharge Records

This report documents the data-quality issues found in the raw patient admission/discharge dataset and the cleaning steps applied to resolve them, as recorded in the workbook's Raw Dataset, Cleaning Process, and Clean Dataset sheets.

1. Dataset Overview

The workbook tracks hospital admission and discharge records (patients, wards, sponsors, admission/discharge timestamps, length of stay, discharge outcome, district, and region), predominantly for facilities in the Kigoma region.

Metric	Value
Raw records (Raw Dataset sheet)	907 rows × 18 columns
Records carried through cleaning workflow	906 rows analyzed
Final clean records (Clean Dataset sheet)	905 rows × 12 columns
Sheets in workbook	Raw Dataset · Cleaning Process · Clean Dataset

Table 1. High-level dataset counts.

Raw Dataset — sample

A representative excerpt of the untouched source data:

Patient Name	Patient Number	Age	Gender	District	Ward	Sponsor	Date Admitted	Date Discharge	Days	Discharge Reason
Babyofmadelina Obed Bulikonyo	92772	1 years	Male	Uvinza	NEONATAL WARD (WARD 2B)	CASH	2024-12-23 01:09:33	2026-01-09 18:48:22	382	Normal Discharge
Babyof Kashindi Philmon	93921	1 years	Female	Manisipaa ya kigoma ujiji	NEONATAL WARD (WARD 2B)	CASH	2025-01-05 15:15:31	2026-01-09 18:49:02	369	Normal Discharge
Mary Japhet Lucas	99223	33 years	Female	Manisipaa ya kigoma ujiji	LABOUR WARD (WARD 1)	CASH	2025-02-28 20:44:11	2026-01-09 10:54:20	314	Normal Discharge
Baraka Abdul Hassan	99453	1 years	Male	Manisipaa ya kigoma ujiji	NEONATAL WARD (WARD 2B)	CASH	2025-03-03 23:43:12	2026-01-09 18:50:23	311	Normal Discharge
Enjoy Sumaili Japhet	99888	21 years	Female	Manisipaa ya kigoma ujiji	LABOUR WARD (WARD 1)	CASH	2025-03-09 06:39:50	2026-01-09 10:57:27	306	Normal Discharge
Helina Esroni Esroni	101331	29 years	Female	Kigoma Vijijini	LABOUR WARD (WARD 1)	CASH	2025-03-24 20:28:38	2026-01-09 11:04:26	290	Escape
Salma Rashid Selemani	38859	32 years	Female	Manisipaa ya kigoma ujiji	LABOUR WARD (WARD 1)	CASH	2025-03-28 10:22:30	2026-01-09 10:54:56	287	Normal Discharge
Fedha Sud Juma	101662	35 years	Female	Manisipaa ya kigoma ujiji	LABOUR WARD (WARD 1)	CASH	2025-03-28 18:34:57	2026-01-09 10:58:03	286	Normal Discharge

Screenshot 1. Raw Dataset sheet (first rows, key columns).

2. Data Quality Issues Identified

The Cleaning Process sheet runs each raw field through validation and standardization formulas, flagging problems column by column.

Field	Issue Found	Detail
Patient Number	Duplicate IDs	44 of 906 rows share a Patient Number with another row (flagged "Duplicate"); 862 are unique
Age	Mixed units in one field	Stored as free text such as "1 years", "11 months" — not usable for numeric analysis
Gender	Inconsistent labels	Checked for values outside "Male"/"Female"; all 906 rows passed as Valid
District	Spelling/naming variants	11 raw spellings collapsed to 10 standardized names (e.g. "Manisipaa ya kigoma ujiji" → "Kigoma Ujiji Mc"; "Buhingwe" → "Buhigwe")

Ward	Free-text ward/room labels	Long raw labels (e.g. "FEMALE SURGICAL & GYNECOLOGICAL WARD (WARD 5)") mapped to 11 standardized ward categories
Sponsor	Unstandardized categories	UNIQUE() inventory of column M surfaced 9 distinct payer categories (CASH, NHIF, NSSF, MHIS, FAST TRACK, CHF, OTHER INSURANCE, STRATEGIES, and one facility code)
Date Admitted	Text, not real dates	904 of 906 rows stored the timestamp as text ("Text Error"), blocking date math and sorting
Date Discharge	Mostly valid	Already numeric/date in nearly all rows; a standardized "Clean Discharge" column was produced
Days (length of stay)	Checked for errors	Row-by-row check: validated as numeric and non-negative — all 906 rows OK
Discharge Reason	Category inventory	UNIQUE() inventory of column W surfaced 5 categories in use: Normal Discharge, Escape, DAMA, Death, Referred
Region	Outlier values	UNIQUE() inventory of column Y surfaced 3 distinct regions: Kigoma (904 rows), Katavi (1), Tabora (1) — the 2 non-Kigoma rows were flagged for review

Table 2. Issues identified per field.

3. Cleaning Methodology

Each field was cleaned with a dedicated formula placed in an adjacent helper column (headers highlighted in green), so the original raw values are preserved side-by-side with the corrected values. Below, each formula is shown in its own box, with the plain-language explanation directly underneath it — so you can read what a formula literally does, and what it means, as two separate things.

Two different validation techniques appear in the workbook:

- Row-by-row checks (Patient Name, Patient Number, Age, Gender, Days) evaluate every single row and return a per-row verdict, e.g. "OK" / "Text Error" / "Negative".
- Category-inventory checks (Sponsor, Discharge Reason, Region) use a single UNIQUE() array formula placed once in row 2 that spills down and lists every distinct value found anywhere in that column, rather than flagging individual rows.

Name, ID, age, and gender checks

Patient Name — name-completeness check (column B)

```
=IF(LEN(TRIM(A2))-LEN(SUBSTITUTE(TRIM(A2)," ",""))>=1,TRIM(A2),"⚠ Only 1 name: "&TRIM(A2))
```

Trims extra spaces and checks the name contains at least one space (i.e. two or more name parts). If not, it prepends a warning flag rather than silently passing a single-word name through.

Patient Number — duplicate check (column D)

```
=IF(COUNTIF($C$2:$C$905,C2)>1,"Duplicate","Unique")
```

Counts how many times this Patient Number appears in the whole column. If more than once, marks the row "Duplicate"; otherwise "Unique". 44 of 906 rows came back Duplicate.

Age — text-to-number conversion (column F)

```
=IF(ISNUMBER(SEARCH("month",E2)), LEFT(E2,FIND(" ",E2)-1)&"/12",
VALUE(LEFT(E2,FIND(" ",E2)-1)))
```

Pulls the leading number out of free text like "1 years" or "11 months". If the text says "month", it's expressed as a fraction of a year (e.g. "11 months" → 11/12); otherwise the number is taken as whole years.

Gender — validity check (column H)

```
=IF(OR(TRIM(G2)="Male", TRIM(G2)="Female"), "Valid", "Inconsistent")
```

Flags anything other than exactly "Male" or "Female" as "Inconsistent". All 906 rows passed as Valid.

Patient Name	Patient Name	Patient Number	Patient Number	Age	Age (Yrs)	Gender	Validity Check
Babyofmadelina Obed Bulikonyo	Babyofmadelina Obed Bulikonyo	92772	Unique	1 years	1	Male	Valid
Babyof Kashindi Philmon	Babyof Kashindi Philmon	93921	Unique	1 years	1	Female	Valid
Mary Japhet Lucas	Mary Japhet Lucas	99223	Unique	33 years	33	Female	Valid
Baraka Abdul Hassan	Baraka Abdul Hassan	99453	Unique	1 years	1	Male	Valid
Enjoy Sumaili Japhet	Enjoy Sumaili Japhet	99888	Unique	21 years	21	Female	Valid
Helina Esroni Esroni	Helina Esroni Esroni	101331	Unique	29 years	29	Female	Valid

Screenshot 2. Cleaning Process sheet — name, ID, age, and gender columns (before → after).

District, ward, and sponsor standardization

District — spelling standardization (column J)

```
=SUBSTITUTE(SUBSTITUTE(TRIM(CLEAN(I2)),"Manisipaa ya kigoma ujiji","Kigoma
Ujiji Mc"),"Buhingwe","Buhigwe")
```

CLEAN() strips non-printable characters, TRIM() removes extra spaces, then two nested SUBSTITUTE() calls rewrite the two known misspellings to one standard name each.

Ward — keyword mapping (column L)

```
=IFS(
  ISNUMBER(SEARCH("ICU",K2)), "ICU",
  ISNUMBER(SEARCH("NEONATAL",K2)), PROPER("NEONATAL"),
  ISNUMBER(SEARCH("LABOUR",K2)), PROPER("LABOUR"),
  ISNUMBER(SEARCH("PAEDIATRIC",K2)), PROPER("PAEDIATRIC"),
  ... (11 conditions total),
  TRUE, "Unmapped: " & K2)
```

Searches the raw ward/room text for a keyword (e.g. "NEONATAL", "LABOUR", "ICU") and returns a short, standardized ward category for the first keyword it matches. A final TRUE branch would label anything unrecognized as "Unmapped" — none occurred, so all 906 rows matched one of 11 categories.

Sponsor — category inventory (column N)

```
=UNIQUE(M2:M907)
```

Lists every distinct payer/sponsor value found anywhere in the raw Sponsor column, spilling down from this one cell. Surfaced 9 categories, letting a reviewer eyeball the list for stray or misspelled entries.

District	District	Ward	Ward	Sponsor	Validity Check
Uvinza	Uvinza	NEONATAL WARD (WARD 2B)	Neonatal	CASH	CASH
Manisipaa ya kigoma ujiji	Kigoma Ujiji Mc	NEONATAL WARD (WARD 2B)	Neonatal	CASH	FAST TRACK
Manisipaa ya kigoma ujiji	Kigoma Ujiji Mc	LABOUR WARD (WARD 1)	Labour	CASH	MHIS
Manisipaa ya kigoma ujiji	Kigoma Ujiji Mc	NEONATAL WARD (WARD 2B)	Neonatal	CASH	NHIF

Screenshot 3. Cleaning Process sheet — district, ward, and sponsor columns.

Date and length-of-stay checks

Date Admitted — text-vs-date check (column P)

```
=IF(ISNUMBER(O2), "Valid Date", "Text Error")
```

A true Excel date is stored internally as a number; text that only looks like a date is not. This returns "Text Error" whenever the raw cell isn't a real number — true for 904 of 906 rows.

Date Admitted (parsed) — re-check (column R)

```
=IF(ISNUMBER(Q2), "Valid Date", "Text Error")
```

The same check applied to a parsed working copy of the date, to confirm the conversion actually produced a real date value.

Days — non-numeric / negative check (column V)

```
=IF(NOT(ISNUMBER(U2)), "Text Error", IF(U2<0, "Negative", "OK"))
```

Checks every row individually: "Text Error" if length of stay isn't a number, "Negative" if it's below zero, otherwise "OK". All 906 rows returned OK.

Date Admitted	Validity Check	Date Admitted	Validity Check	Date Discharge	Clean Discharge	Days	Days
2024-12-23 01:09:33	Text Error	12/23/2024 1:09	Valid Date	2026-01-09 18:48:22	1/9/2026 18:48	382	OK
2025-01-05 15:15:31	Text Error	1/5/2025 15:15	Valid Date	2026-01-09 18:49:02	1/9/2026 18:49	369	OK
2025-02-28 20:44:11	Text Error	2/28/2025 20:44	Valid Date	2026-01-09 10:54:20	1/9/2026 10:54	314	OK
2025-03-03 23:43:12	Text Error	3/3/2025 23:43	Valid Date	2026-01-09 18:50:23	1/9/2026 18:50	311	OK
2025-03-09 06:39:50	Text Error	3/9/2025 6:39	Valid Date	2026-01-09 10:57:27	1/9/2026 10:57	306	OK

Screenshot 4. Cleaning Process sheet — Date Admitted, Date Discharge, and Days validation.

Discharge Reason and Region checks

Discharge Reason — category inventory (column X)

```
=UNIQUE(W2:W907)
```

Spills the 5 distinct discharge reasons found in the raw data (Normal Discharge, Escape, DAMA, Death, Referred) into column X for a quick visual scan — it does not flag individual rows.

Region — category inventory (column Z)

```
=UNIQUE(Y2:Y907)
```

Spills the 3 distinct region values found in the raw data (Kigoma, Katavi, Tabora) into column Z. This is how the 2 non-Kigoma rows were spotted — anything other than "Kigoma" stands out immediately in the short spilled list.

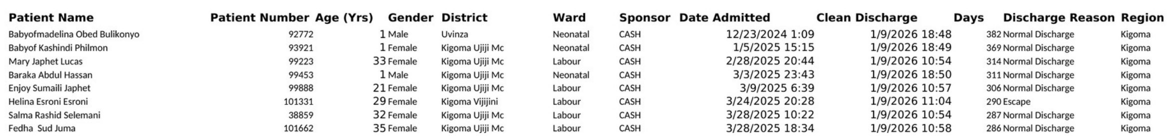
Date column fix applied: Text to Columns

The broken Date Admitted column (text instead of real dates) was standardized using Excel's Text to Columns feature, following these steps:

- 1. Select the entire Date column (Column O).
- 2. Go to the Data tab and click Text to Columns.
- 3. Choose Delimited → Next.
- 4. Uncheck all delimiter boxes → Next.
- 5. Under Column data format, choose Date and select YMD.
- 6. Click Finish.
- 7. Format the column as Short Date or a custom YYYY-MM-DD HH:MM:SS format (Ctrl+1).

4. Final Clean Dataset

The Clean Dataset sheet keeps only the corrected values, condensed to 12 columns: Patient Name, Patient Number, Age (Yrs), Gender, District, Ward, Sponsor, Date Admitted, Clean Discharge, Days, Discharge Reason, and Region.



Patient Name	Patient Number	Age (Yrs)	Gender	District	Ward	Sponsor	Date Admitted	Clean Discharge	Days	Discharge Reason	Region
Babyofmadelina Obed Bulikonyo	92772	1	Male	Uvinza	Neonatal	CASH	12/23/2024 1:09	1/9/2026 18:48	382	Normal Discharge	Kigoma
Babyof Kashindi Philmon	93921	1	Female	Kigoma Ujiji Mc	Neonatal	CASH	1/5/2025 15:15	1/9/2026 18:49	369	Normal Discharge	Kigoma
Mary Japhet Lucas	99223	33	Female	Kigoma Ujiji Mc	Labour	CASH	2/28/2025 20:44	1/9/2026 10:54	314	Normal Discharge	Kigoma
Baraka Abdul Hassan	99453	1	Male	Kigoma Ujiji Mc	Neonatal	CASH	3/3/2025 23:43	1/9/2026 18:50	311	Normal Discharge	Kigoma
Enjoy Sumaili Japhet	99888	21	Female	Kigoma Ujiji Mc	Labour	CASH	3/9/2025 6:39	1/9/2026 10:57	306	Normal Discharge	Kigoma
Helina Esroni Esroni	101331	29	Female	Kigoma Vijijini	Labour	CASH	3/24/2025 20:28	1/9/2026 11:04	290	Escape	Kigoma
Salma Rashid Selemani	38859	32	Female	Kigoma Ujiji Mc	Labour	CASH	3/28/2025 10:22	1/9/2026 10:54	287	Normal Discharge	Kigoma
Fedha Sud Juma	101662	35	Female	Kigoma Ujiji Mc	Labour	CASH	3/28/2025 18:34	1/9/2026 10:58	286	Normal Discharge	Kigoma

Screenshot 5. Clean Dataset sheet — ready for analysis.

5. Validation Summary

After running every check, each flagged field was reviewed and closed out as follows:

Field	Validation Outcome
Date Admitted	Confirmed broken: 904 of 906 rows were stored as text instead of real dates. Standardized using Excel's Text to Columns feature (Delimited → Date → YMD), converting the column to true date/time values.
Days (length of stay)	Confirmed valid: all 906 rows are numeric and non-negative. No correction needed.
Discharge Reason	Confirmed valid: all 5 categories found (Normal Discharge, Escape, DAMA, Death, Referred) are legitimate, expected values. No correction needed.
Region	Confirmed valid: all 3 values found (Kigoma, Katavi, Tabora) are legitimate entries, not typos. No correction needed.
Patient Number	44 duplicate Patient Numbers confirmed as readmitted patients — the same patient returning for a separate admission episode — not data-entry duplicates. Retained in the dataset as distinct admission records.

Table 3. Final validation outcome per flagged field.

6. Summary & Recommendations

- The 44 duplicate Patient Numbers are confirmed readmissions, not errors — keep them as separate admission records, but be aware when counting "patients" vs. "admissions" that these represent the same person more than once.
- Date Admitted has been standardized via Text to Columns; re-apply the same check to Date Discharge if it is ever re-entered as text.
- Days, Discharge Reason, and Region all passed validation as-is — no further correction needed for these fields.
- Retain the standardized District, Ward, and Discharge Reason categories for consistent reporting/analysis going forward.
- The Clean Dataset sheet is ready to be used as the analysis-ready source for downstream reporting.